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1. $V_{t0} = .45V$, $V_{psat} = .6V$, $k_n' = 115 \mu A/V^2$, $\gamma_n = 0$, $\lambda_n = 0$

$2|\phi_F| = .6V$, $\phi_0 = .65V$, $t_{ox} = 5nm$, $m_j = .5$, $m_{jsw} = .44$, $x_d = .05 \mu m$

$C_{j0} = 2(10^{-3}) F/m^2$, $C_{jsw0} = 2.75 \times 10^{-10} F/m$, $W = 2 \mu m$, $L = .25 \mu m$

$L_D = L_S = .8 \mu m$, $V_{DD} = 2.5$

a. $V_{GS} = 2.5$; $V_{DS} = 2.5, .5$, and 0

C_D means diff cap btw

$V_{DS} = V_{GS} = 2.5$

velo sat

$\epsilon_{ox} = 3.455 \times 10^{-11} F/m$

$C_{ox} = \frac{3.455 \times 10^{-11}}{t_{ox} = 5 \times 10^{-9}} = .00691$

$C_G = \frac{2}{3} C_{ox} W L_{eff} + 2 C_0 W$

$L_{eff} = L - 2x_d = .15 \mu m$

$C_0 = C_{ox}(x_d) = 3.455 \times 10^{-10}$

$\boxed{C_G = 2.764 \times 10^{-15} F}$

$C_D = C_j L_S W + C_{jsw}(2L_S + W)$

$C_j = \frac{C_{j0}}{(1 + \frac{V_D}{\phi_0})^{m_j}} = \frac{2 \times 10^{-3}}{(1 + \frac{2.5}{.65})^{.5}} = 9.085 \times 10^{-4}$

$C_{jsw} = \frac{C_{jsw0}}{(1 + \frac{V_D}{\phi_0})^{m_{jsw}}} = \frac{2.75 \times 10^{-10}}{(1 + \frac{2.5}{.65})^{.44}} = 1.373 \times 10^{-10}$

$9.085 \times 10^{-4} (.8)(2) 10^{-12} +$

$1.373 \times 10^{-10} (2(.8) + 2) 10^{-6}$

$= \boxed{1.948 \times 10^{-15} F = C_D}$

$V_{DS} = .5$

linear

$C_G = .00691)(2)(.15) 10^{-12} + 2(.05)(.00691)$

$(2)(10^{-12})$

$\boxed{C_G = 3.455 \times 10^{-15} F}$

$C_j = \frac{2 \times 10^{-3}}{(1 + \frac{.5}{.65})^{.5}} = .0015$

$C_{jsw} = \frac{2.75 \times 10^{-10}}{(1 + \frac{.5}{.65})^{.44}} = 2.139 \times 10^{-10}$

$A = 1.6 \times 10^{-12}$

$P = 3.6 \times 10^{-6}$

C_D

$\boxed{C_j A + C_{jsw} P = 3.176 \times 10^{-15} F}$

$V_{DS} = 0$

linear

same C_G as $V_{DS} = .5$

$\boxed{C_G = 3.455 \times 10^{-15} F}$

$C_j = \frac{.002}{1^{.5}} = .002$

$C_{jsw} = 2.75 \times 10^{-10}$

$\boxed{C_j A + C_{jsw} P = 4.19 \times 10^{-15} F = C_D}$

b.

C_g for all V_{DS}:

Same equation as linear,

So C_g for all V_{DS} 2.5 to 0 = 3.455×10^{-15} F

The diffusion capacitances also don't change from part a.

$V_{DS} = 2.5$	$V_{DS} = .5$	$V_{DS} = 0$
$C_D = 1.948 \times 10^{-15}$ F	$C_D = 3.176 \times 10^{-15}$ F	$C_D = 4.19 \times 10^{-15}$ F

2.

a. $K_n' = 115$ $V_{TN0} = .43$, $V_{Tbody} = .6$, $V_{Dsat} = .63$, $\lambda = 0$

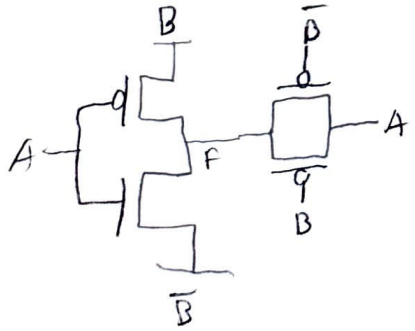
$K_p' = 30$ $V_{TP0} = -.4$, $V_{Tbody} = -.55$, $V_{Dsat} = -1$, $\lambda = 0$

$V_{DD} = 2.5$

A	B	M1	M2	M3	M4	F (voltage)	F (logic)
0	0	off	on	on	on	0	0
0	1	off	on	off	off	2.5	1
1	0	on	off	on	on	2.5	1
1	1	on	off	off	off	0	0

b. XOR GATE!!

$A\bar{B} + \bar{A}B = F$

$\overline{\bar{A}\bar{B} + AB} = F$

c. Logic swing is 2.5 V. Swings from 2.5 to 0.